

A detailed illustration of two birds of prey, possibly vultures or eagles, perched on a gnarled, light-colored branch. The bird in the upper right is shown in profile, facing left, with its head slightly turned. It has dark brown plumage with lighter, scaly patterns on its wings. The bird in the lower center is facing forward, looking slightly to the left. It has a lighter, more mottled brown and white plumage. Both birds have sharp, hooked beaks and powerful talons gripping the branch. The background is plain white.

ORNITHOLOGICAL MASTERCLASS

37 AVIAN OLFACTION

It's time to bust a long-standing myth: **birds can smell** – and they use their noses more than you'd expect.

Marc Gilles tells us about the surprising ways birds use olfaction in their daily lives. Ready to take a nosedive into the scented world of birds?

The world is full of odours, and many of them matter. Odours can signal the presence of food, disease, predators or potential partners, all of which is quite useful. The sense of smell, or olfaction, is one of the most ancient, found in all organisms from bacteria to plants and animals. Yet, there is one taxon widely believed to have little or no sense of smell: birds. Why would birds be the only organisms that cannot smell?

The origin of the myth

The myth that birds can't smell is thought to have originated in the 1820s. At the time, it was widely accepted that Turkey Vultures use smell to locate carrion. However, the famed ornithologist and painter John James Audubon thought differently. He conducted field experiments to show that vultures use vision, not olfaction, to find food. For example, he placed a painting of a dead sheep in the field which successfully attracted vultures, thus 'proving' that they rely on visual cues. The experiments were later found to be flawed, but the damage was done – Audubon had achieved his goal of “exploding” (his words) the idea that birds use smell. The myth took off, and remarkably, it still lingers today – even among ornithologists.

A PERSISTENT MYTH

It's not all on Audubon though. Several other factors helped make the misbelief stick. First, anatomy: many birds – especially songbirds – have relatively small olfactory bulbs (the brain structures involved in processing smells). Second, behaviour: birds don't look like they're sniffing. Unlike mammals that wiggle their wet noses or insects that wave their antennae, birds have rigid beaks and nostrils that give little away, and they don't engage in obvious scent-marking. Third, theory: birds have

evolved developed visual and auditory systems, alongside elaborate plumage and complex vocalisations. Some researchers have speculated that this came at the cost of olfaction.

Even songbirds with their small olfactory bulbs have surprisingly high olfactory acuity

THE AVIAN NOSE

Despite the myth's persistence, researchers began to question it in the 1960s. Anatomical studies first showed that birds have, in fact, a fully equipped olfactory system. Indeed, the 'nose' of birds is similar to that of many other vertebrates. Air enters through a pair of nostrils into internal nasal cavities, where it is filtered, warmed, moistened and chemically sampled by olfactory receptors. The resulting chemical information is then transmitted via olfactory nerves to the olfactory bulb in the brain, where it is processed. Electrophysiological studies then showed that birds – even songbirds with their small olfactory bulbs – have surprisingly high olfactory acuity, comparable to that of rats! So, yes, birds can smell – and quite well.

TURKEY VULTURE: PUBLIC DOMAIN, BLUE TIT: ALLAN DREWITT/BJO

How to test avian olfaction?

SATELLITE TELEMETRY

Birds are rendered anosmic (unable to smell) by nostril plug, zinc-sulphate nasal washing, or sectioning the olfactory nerve, and equipped with GPS tags. Their ability to return to their home site or migratory route after displacement, or to locate food sources, is compared to control birds. If olfaction guides navigation or foraging, anosmic birds show impaired orientation, as found in pigeons and petrels.

Y-MAZE TRIALS

In two-choice tests, birds are released into a Y-maze with different odours at each arm. Consistent preference for a specific odour indicates olfactory discrimination. This method has shown that petrels can recognise the scent of their partner, nest, and eggs.

BEHAVIOURAL ASSAYS

Birds are exposed to test odours and their reactions are recorded. For example, Zebra Finch chicks increase begging in response to parental scent, sleeping King Penguins wake up in response to conspecific scent, and Blue Tits avoid their nestbox if it contains predator scent.



◀ Blue Tits use smell to select aromatic plants to furnish their nests and create a healthy environment for their chicks.

What do birds use smell for?

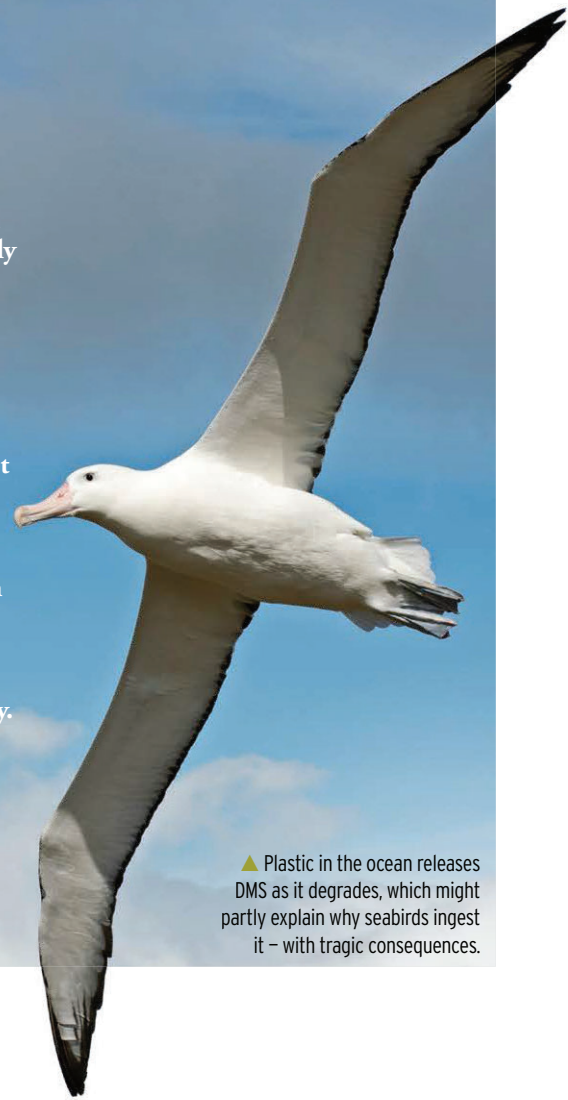
The smell of food

Imagine you're a seabird, soaring over the vast open ocean, in search of your next meal. Somewhere below, food-rich zones teem with krill and fish, but how can you find them? Albatrosses and petrels have a trick – they smell them. In upwelling zones, phytoplankton gathers and is eaten by krill and fish, releasing a chemical called dimethyl sulfide (DMS) into the air. These DMS plumes act as olfactory landmarks (or rather seamarks), which albatrosses and petrels can detect from kilometres away. The ocean surface, once thought to be featureless, is actually full of invisible, smelly cues.

Something similar occurs on land. You know the smell of freshly cut grass? White Storks know it too, and to them, it means one thing – mealtime. The prey of storks (snails, frogs and small rodents) live in tall grass, so once the grass is mowed,

it is easier to find and catch them. Freshly mown fields are like all-you-can-eat buffets for storks. Storks that are located downwind from the fields – even kilometres away – smell them and quickly come to feast.

And after Audubon, we need to set the record straight. Several studies have now clearly shown that Turkey Vultures do use odours to find food. Vultures are especially attracted to the sulphur-based chemicals released by decaying flesh. Plot twist – similar chemicals are found in some oil paints, so it may have been the smell of the painting, not the image of the dead sheep, that lured the vultures in Audubon's experiment! Fun fact – in the 1930s, the Union Oil Company added those same chemicals into their gas lines to detect leaks in the remote backcountry. Circling vultures meant a leak!



▲ Plastic in the ocean releases DMS as it degrades, which might partly explain why seabirds ingest it – with tragic consequences.

The smell of home

In the 1970s, scientists were testing whether homing pigeons rely on visual cues from the horizon to find home, and they made a surprising discovery. Pigeons raised behind glass screens (with a view but no wind) struggled to navigate back home, while those raised behind an

opaque palisade (no view but exposed to wind) returned home easily. The scientists concluded, “it must be something in the air!” A curious idea emerged – what if pigeons used smell to navigate? Today, after over 50 years of extensive research and fierce debates, it is clear that olfaction is crucial for the navigation of homing pigeons. They develop an ‘olfactory map’ of the region around their home aviary by sniffing windborne scents, which they can use to navigate over long distances.

Migratory bird species may also use olfactory cues as a compass. An experiment on Grey Catbirds, a songbird that migrates between North and Central America, suggests so. Adult catbirds that were rendered anosmic can't orient as well as

control adults (with full olfactory abilities) and instead fly in the wrong direction.

For petrels, home is a burrow tucked in the middle of a colony of thousands of nearly identical burrows. Furthermore, petrels usually return to their home burrow under the cover of night, without calling, to avoid predation. So, how do they find their burrow? By smell, of course. Nocturnal petrels like Leach's Petrel and Blue Petrel can recognise the unique scent of their own nest. In maze experiments, these birds consistently choose their own burrow's scent over others, while birds rendered anosmic can't make any choice. Even when the nest entrance is moved, Blue Petrels find their nest, proving it's the smell, not the location, that matters.



► Pigeons find it harder to navigate when their right nostril is blocked than their left.

The smell of danger

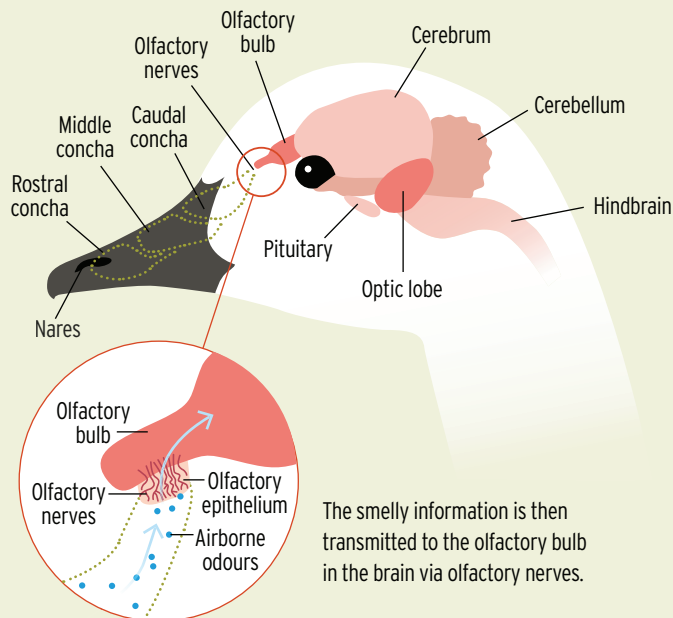
Predators leave behind more than just tracks – they also leave scents. Birds can pick up on these chemical cues and adjust their behaviour to avoid becoming a predator's next meal. Blue Tits, for instance, hesitate before entering their nest box if it smells of Ferret. House Sparrows rush their meals at feeders that are scented with Cat faeces. Interestingly, domestic chickens avoid areas scented with faeces from Tigers and Dholes, which are natural predators of their ancestor, the Red Junglefowl – suggesting that this recognition is innate.

In Rollers, it's not the odour of the predator itself, but an alarm scent from their chicks, that triggers caution. When threatened, chicks vomit a foul-smelling liquid to deter the predator. Parents returning to the nest detect the stench and approach more cautiously, delaying entry – just in case the predator is still lurking inside. Similarly, Red-legged Partridges avoid areas scented with the pungent secretions that Hoopoes release when threatened. In other words, they can eavesdrop on another bird's alarm scent.



THE AVIAN NOSE

Birds possess a fully equipped olfactory system: air enters through nostrils in the nasal cavities. At the caudal concha, which is covered with olfactory epithelium, it is chemically sampled by olfactory receptors.



The smell of love

How do birds find the perfect mate?

Step one: recognise conspecifics In all species tested so far, birds prefer the scent of a member of their own species over that of a different species. This can be especially important for closely related species. For example, Black-capped Chickadees and Carolina Chickadees – two closely related North American songbirds – both prefer the scent of their conspecifics, which may prevent hybridisation.

Step two: discriminate individuals of the opposite sex This might sound simple, but in many bird species where females and males look alike (sexually monomorphic), it's not that easy. Song Sparrows are visually monomorphic, but not chemically – females and males smell different, especially during the breeding season. This olfactory sex difference allows them to tell the sexes apart when it matters most.

Step three: assess genetic quality and compatibility Body odour is linked to genetic makeup, so sniffing a potential mate can help avoid inbreeding and increase offspring fitness. Storm Petrels and Humboldt Penguins, for instance, prefer the scent of non-kin over kin, likely limiting the risk of inbreeding. Song Sparrows prefer the scent of individuals that have a dissimilar genotype at the major histocompatibility complex (MHC),



a key region involved in immune function. This preference may increase the survival of their offspring by enhancing their pathogen resistance.

Step four: evaluate phenotypic quality Bird odours may have evolved as invisible sexual signals, which females and males could use to advertise their quality during mate choice. In Dark-eyed Juncos, scent reflects reproductive success. Females with more 'female-like' odours and males with more 'male-like' odours produce more offspring. In contrast, males with more 'female-like' odours, are more likely to be cuckolded.

Finally, once birds found the right partner, they might want to stick to it. Antarctic Prions and Blue Petrels can recognise their mate by smell, which likely helps them maintain long-term pair bonds.

The smell of family

Recognising kin isn't important just for choosing a mate – it also matters for parenting. In Zebra Finches, young chicks beg more in response to the odour of their parents than to that of foreign adults. Even more strikingly, cross-fostering experiments show that chicks respond more to the scent of their genetic mother, whom they've never met, than their adoptive mum! This suggests that they learn the chemical signatures of their mother before hatching, while still in the egg.

There's evidence that parents, too, can recognise the odour of their offspring. In Zebra Finches, males (but not females, surprisingly) can recognise their chicks' scent. What's more, female Zebra Finches and Blue Petrels can recognise the scent of their eggs.

Smelly birds

Not only do birds smell – as in, they have a sense of smell – they also smell in the sense that they have a body odour. Some species are famously foul smelling, like hoopoes, anis or the Hoatzin (nicknamed the ‘stinkbird’). Birds don’t just stink – Crested Auklets smell like tangerines, petrels have a musky scent and Hooded Pitohuis have a sour perfume. In fact, all birds have a scent. But where does it come from?

PREEN OIL: THE PERFUME OF BIRDS

At the base of a bird’s tail lies a special gland – the uropygial (or preen) gland. During preening, birds use their bill to spread the gland’s waxy secretion (preen oil) over their entire plumage. Preen oil has many functions: it waterproofs feathers, protects them from wear, and is an important source of body odour. It contains a mix of volatile and non-volatile compounds. The latter can break down – physically or with the help of microbes – into additional volatile compounds once applied to the plumage.

Preen oil composition varies by species, sex, individual, life stage, breeding stage, diet, hormone levels, infection status, and even MHC genotype. Therefore, preen oil contains a wealth of information,

► As they preen, birds coat their feathers with oil from the preen gland, much like perfume.



and birds smell to use this information. Indeed, as we’ve seen, some birds can use scent to recognise species, sex or individual identity, and to assess genetic compatibility via MHC dissimilarity.

BIRD SMELL: FEMALE SEXUAL SIGNALS?

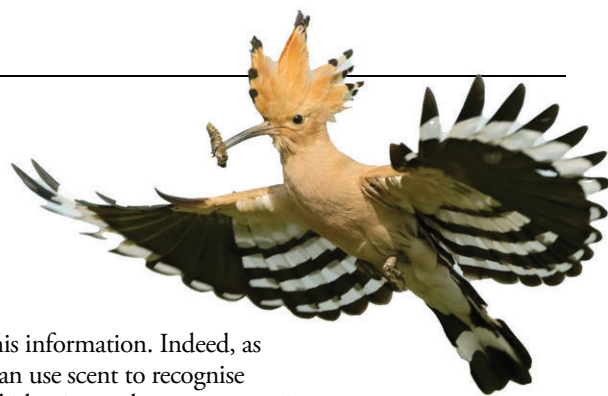
In many species, the preen gland and its secretion show sexual dimorphism. During breeding, females often have a larger uropygial gland, and a greater abundance and diversity of preen-oil compounds (or associated microbes) than at other times. This chemical dimorphism may reflect a greater use of olfactory sexual signals by females than by males. While males often advertise their quality with songs, courtship displays and colourful plumage, females may use a less conspicuous modality – smell.

SMELLY PROTECTION

Bird smell can also serve as a defence against parasites and predators. Some birds reduce their odour to avoid detection. For example, sandpipers that nest on the ground in the high Arctic produce a less odorous preen oil during incubation, chemically camouflaging themselves to reduce the risk of being sniffed out by predators. In contrast, other birds turn up the smell as a deterrent. Hoopoes’ preen oil repels blood-sucking mosquitoes and blackflies, while nesting Green Woodhoopoes secrete foul-smelling preen oil when threatened by predators.

BEYOND PREEN OIL

Preen oil is not the only source of scent. The tangerine smell of Crested Auklets, for instance, comes from specialised feathers. Other sources include powder down, skin, faeces and stomach oils. There’s still a lot to learn about bird odour and how it’s used. ■



▲ The smell of Hoopoes has been likened to that of rotten meat.

▼ The Hoatzin reeks of cow manure because it ferments its food (leaves) in its foregut, much like a cow does.



The women who smelt the truth

The myth of scentless birds didn’t just fade away – it was challenged by pioneering researchers, many of whom were women. This is surprising given that academia has historically been male-dominated. We don’t know the exact gender balance or why this happened, but female researchers have been instrumental in reshaping our understanding of avian olfaction.

Betsy Bang Showed that birds have well-developed olfactory structures.

Bernice Wenzel Used electrophysiology in the 1960s and 1970s to show that birds have a functioning olfactory system.

Gabrielle Nevitt Revealed that seabirds use olfactory cues like DMS to navigate across the open ocean.

Silke Steiger Provided genetic evidence that olfaction matters in avian evolution.

Julie Hagelin Showed the role of odours in sexual selection in birds such as the Crested Auklets

Anna Gagliardo Demonstrated that domestic pigeons rely on smell to navigate

Danielle Whittaker Found that even songbirds like the Dark-eyed Junco use odours for social communication.

Barbara Caspers Uncovered how Zebra Finches use smell for kin recognition.

The author

Marc Gilles is a postdoctoral researcher at Bielefeld University where he investigates the role of odours in birds, from shorebirds to songbirds.

Find out more

Creece *et al.* 2025. *Ibis*. doi: 10.1111/ibi.13398

Gilles, M. 2024. *The role of preen oil and body odour in birds*. PhD thesis, Bielefeld University.

Grieves *et al.* 2022. *Biological Reviews* **97**: 1193–1209.

Whittaker & Hagelin. 2021. *Journal of Chemical Ecology* **47**: 43–62.

Whittaker, D.J. 2022. *The Secret Perfume of Birds: Uncovering the Science of Avian Scent*. Johns Hopkins University Press.